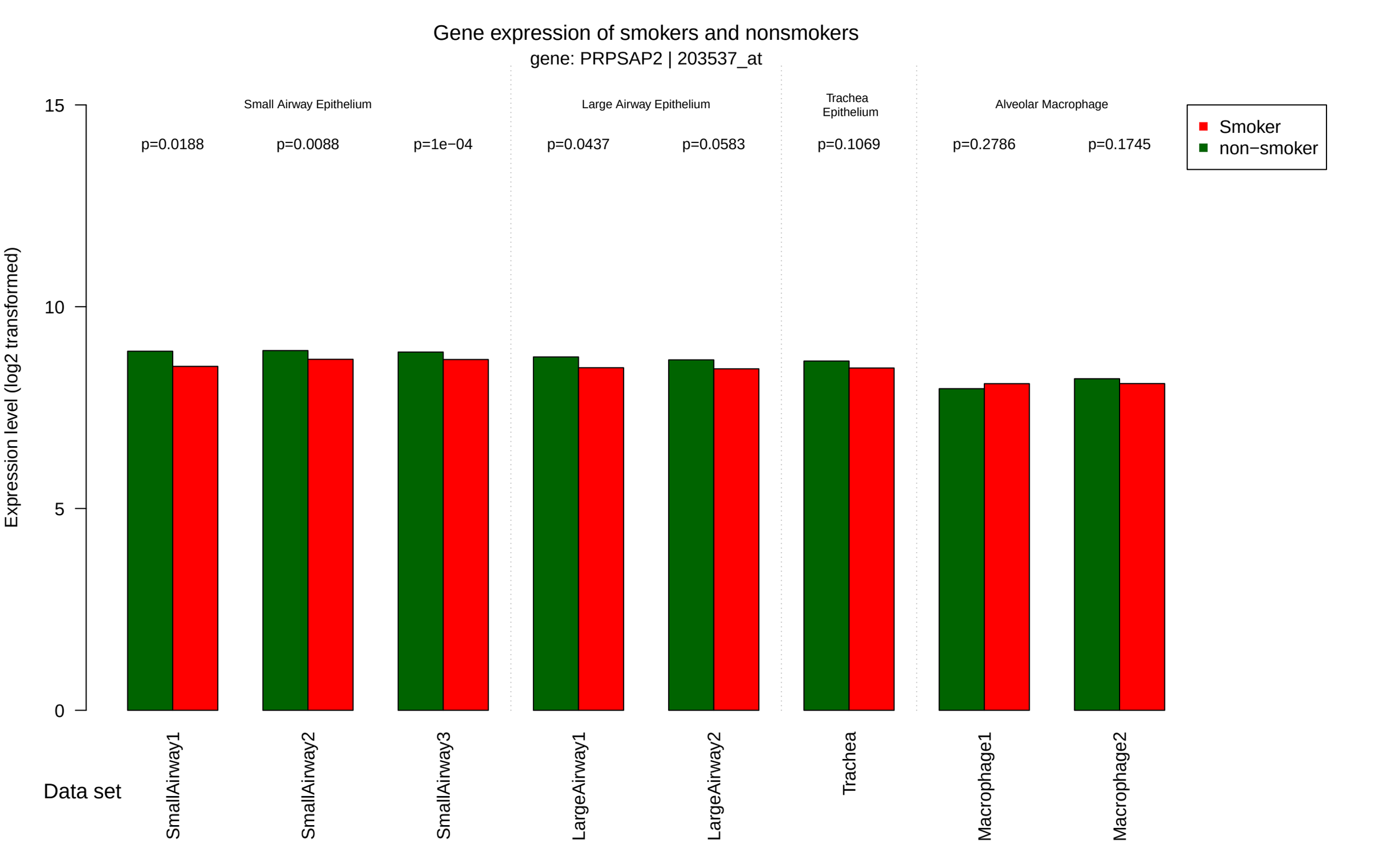
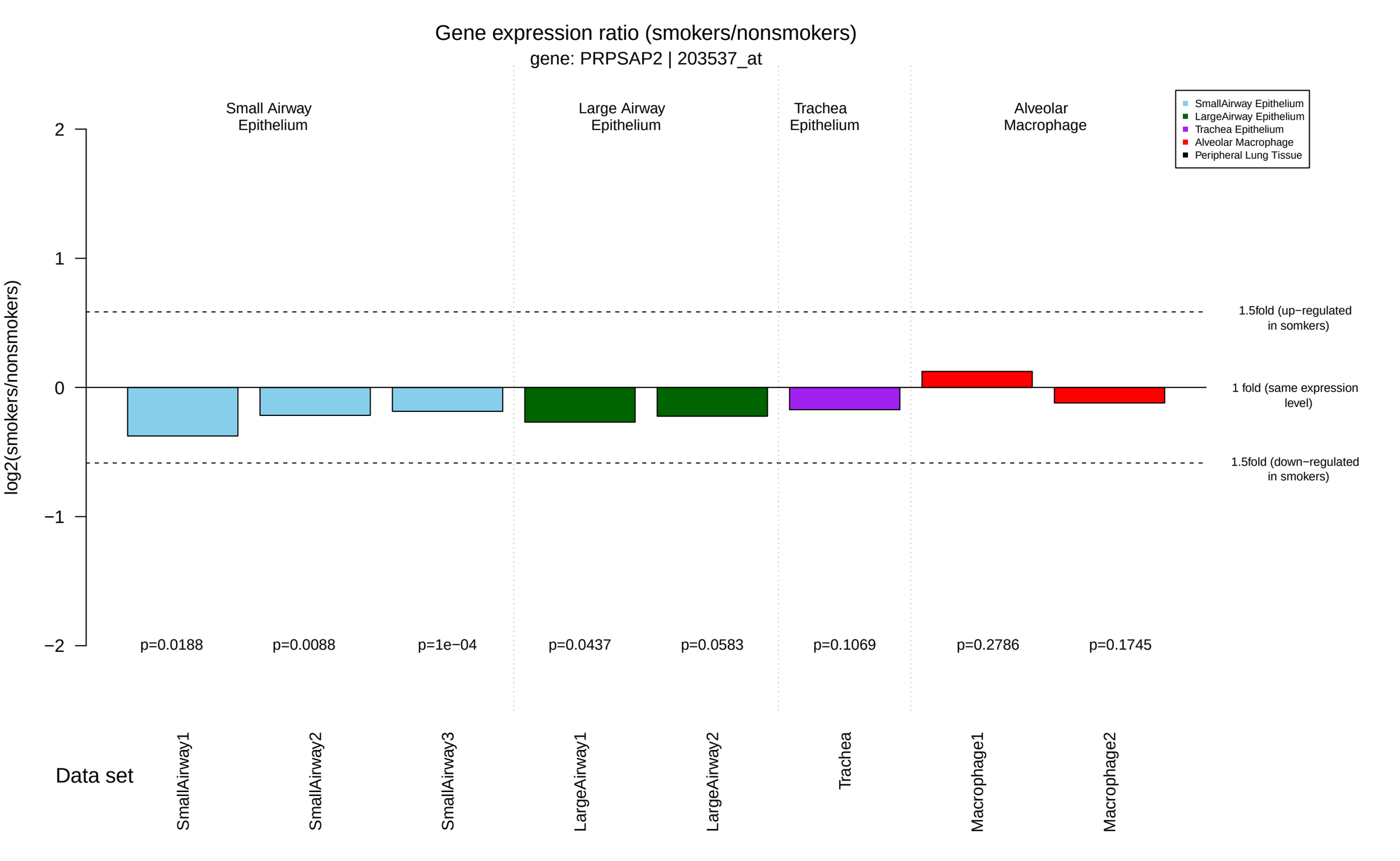
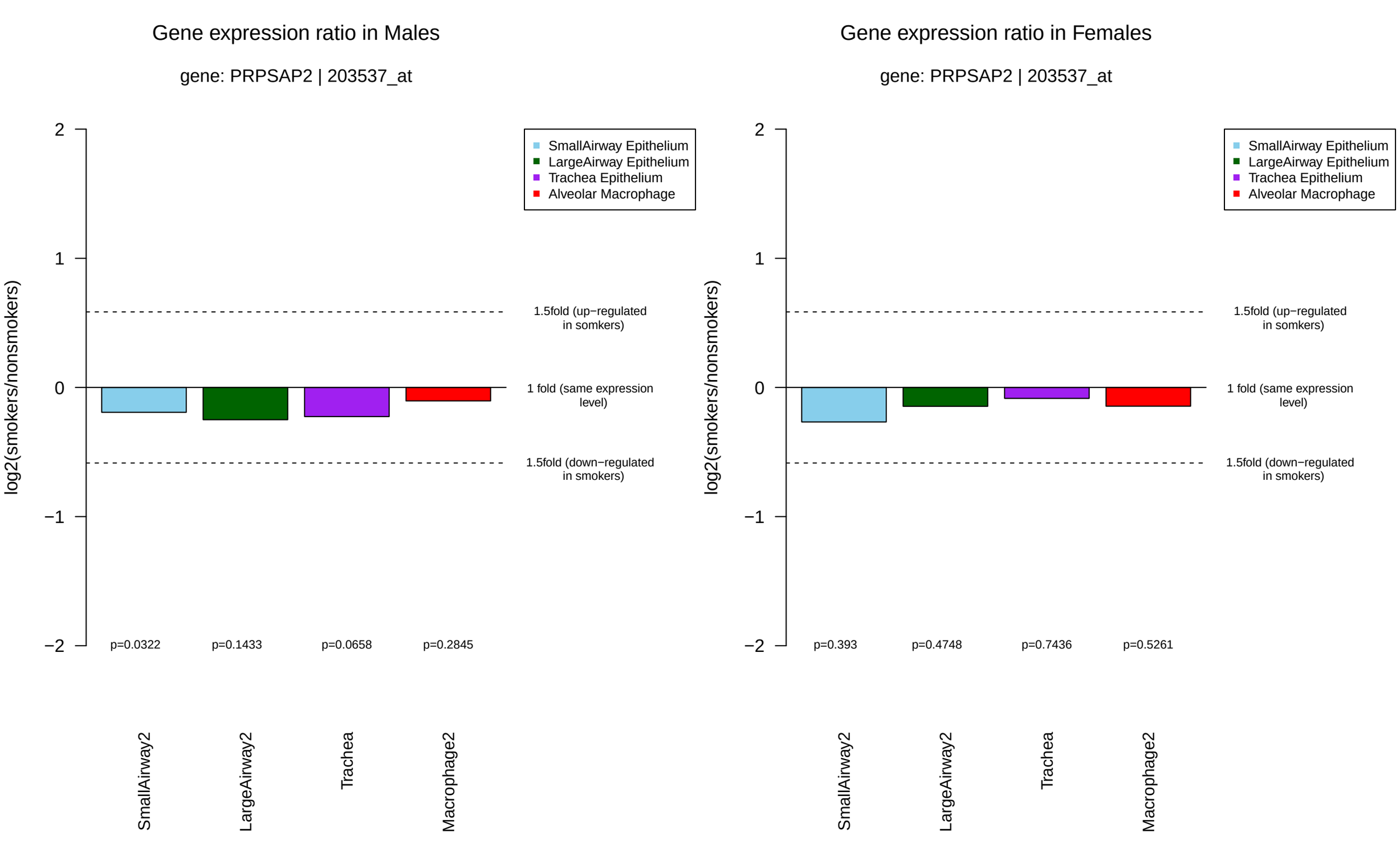
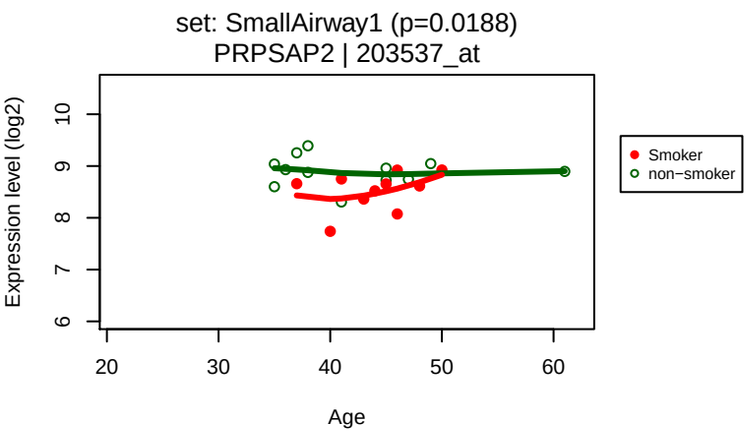
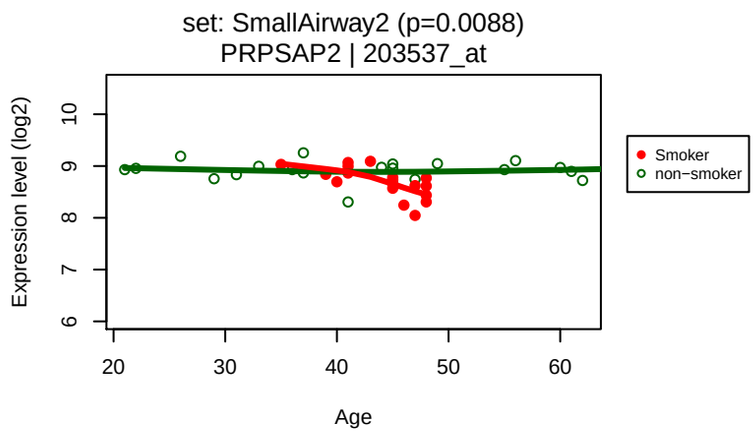
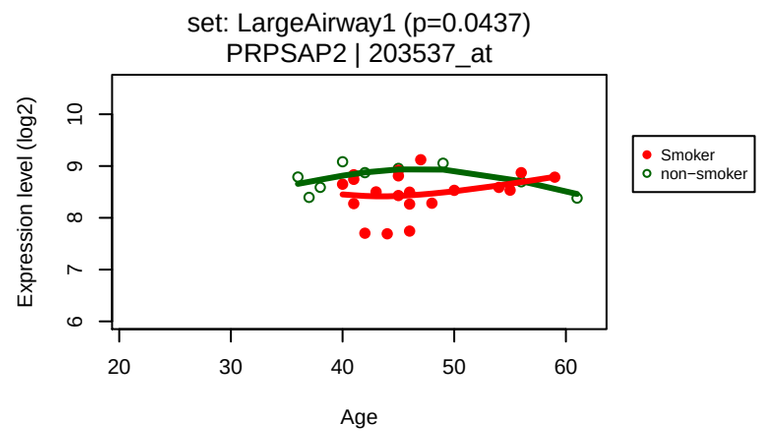

Gene expression of smokers and nonsmokers
gene: PRPSAP2 | 203537_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0188
p=0.0088
p=1e−04
p=0.0437
p=0.0583
p=0.1069
p=0.2786
p=0.1745
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

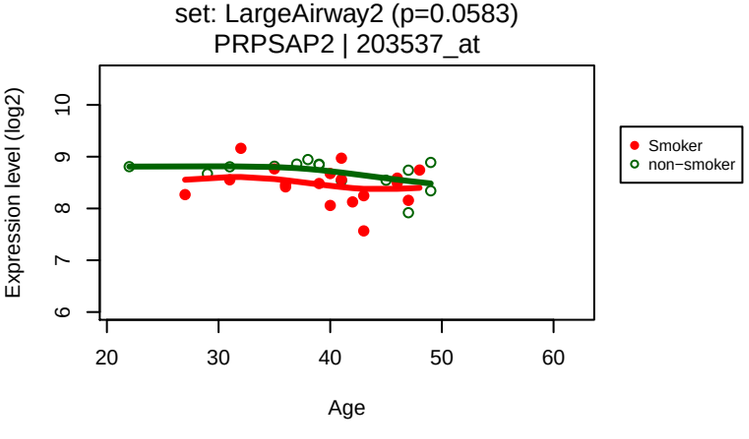

Gene expression ratio (smokers/nonsmokers)
gene: PRPSAP2 | 203537_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0188
p=0.0088
p=1e−04
p=0.0437
p=0.0583
p=0.1069
p=0.2786
p=0.1745
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

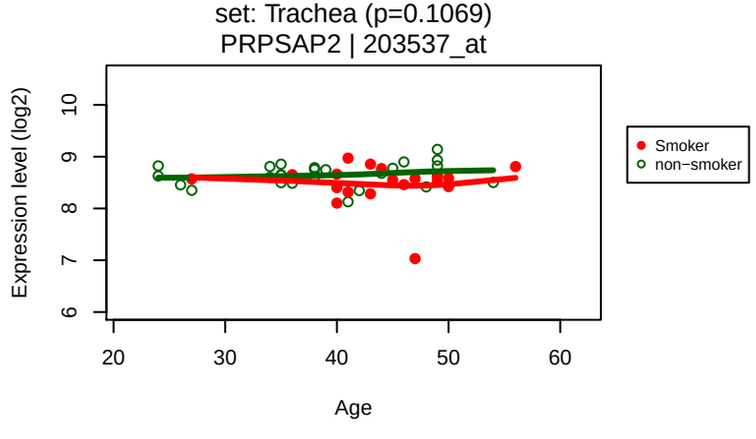

Gene expression ratio in Males
gene: PRPSAP2 | 203537_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0322
p=0.1433
p=0.0658
p=0.2845
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRPSAP2 | 203537_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.393
p=0.4748
p=0.7436
p=0.5261
SmallAirway2
LargeAirway2
Trachea
Macrophage2

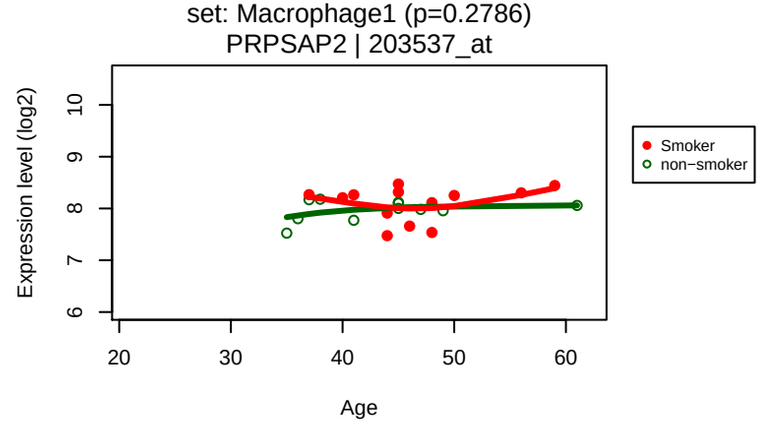

set: SmallAirway1 (p=0.0188)
PRPSAP2 | 203537_at
Expression level (log2)
Age
Smoker
non−smoker

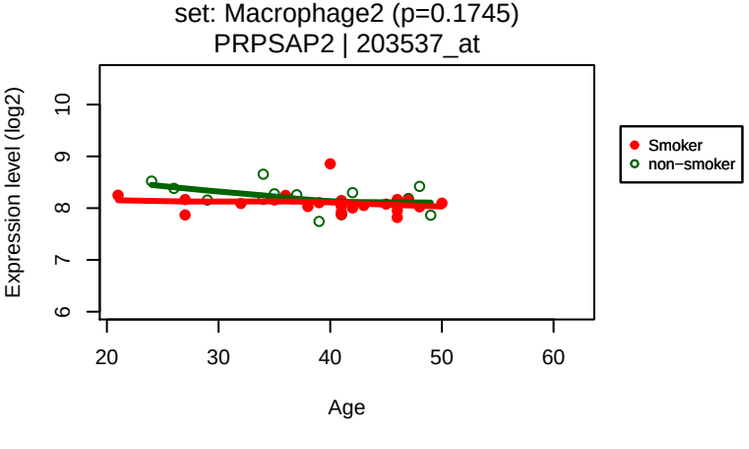

set: SmallAirway2 (p=0.0088)
PRPSAP2 | 203537_at
Expression level (log2)
Age
Smoker
non−smoker

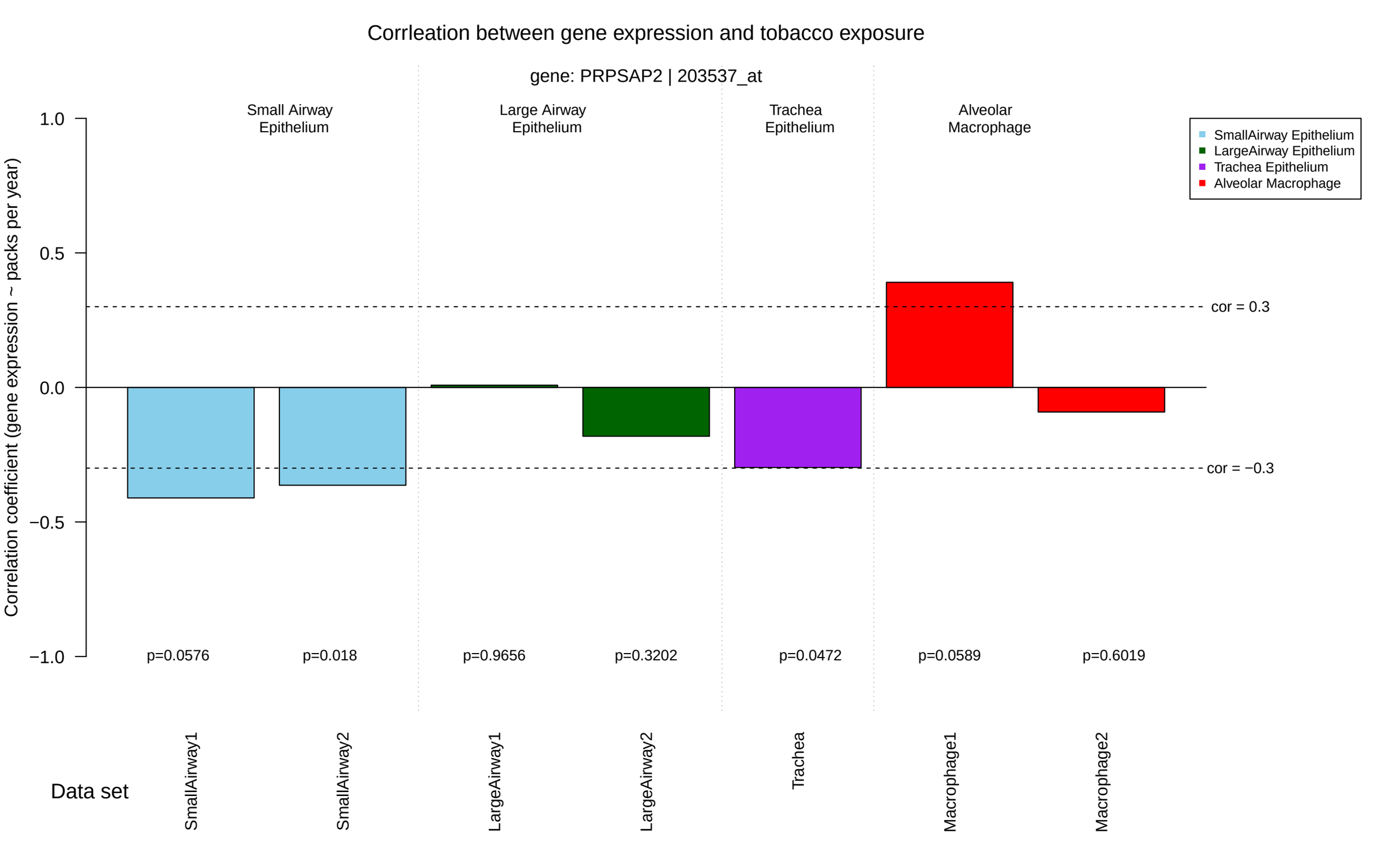

set: LargeAirway1 (p=0.0437)
PRPSAP2 | 203537_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0583)
PRPSAP2 | 203537_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1069)
PRPSAP2 | 203537_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2786)
PRPSAP2 | 203537_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1745)
PRPSAP2 | 203537_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRPSAP2 | 203537_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0576
p=0.018
p=0.9656
p=0.3202
p=0.0472
p=0.0589
p=0.6019
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2